hipTsEnc:1.11

API Reference

X Platform, IP JPEG XS Encoder TS 1.20.5

27/10/2025

appear

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1 Overview

Changelog:

1.11

- **Changed**
  - `hipTsEncoder` from 1.6 to 1.7

1.10

- **Changed**
  - `hipTsEncoder` from 1.5 to 1.6
  - `hipTestGenerator` from 1.0 to 1.1

1.9

- **Changed**
  - `hipEncStatus` from 1.5 to 1.6

1.8

- **Changed**
  - `hipEncTransport` and 'hipTsEncoder' from 1.4 to 1.5

1.7

- **Changed**
  - `hipStatus` from 1.4 to 1.5

1.6

- **Changed**
- The exported package `videoUsabilityInfo` is bumped from 1.0 to 1.1.
- The exported package 'hipTsEncoder' is bumped from 1.3 to 1.4.

1.5

- **Added**
  - `dpi` 1.0
  - `dpiSettings` 1.0
- **Changed**
  - `hipTsSettings` from 1.1 to 1.2
  - `hipTsEncoder` from 1.2 to 1.3

1.4

- **Changed**
  - `hipTsEncoder` from 1.1 to 1.2

1.3

- **Changed**
  - `nmosStatus` from 1.0 to 1.1

1.2

- ** Added **
  * Added `nmosConfig` 1.0
  * Added `nmosRegistryConfig` 1.0;
  * Added `nmosLabelConfig` 1.0;
  * Added `networkManagerProtocols` 1.1.
  * Added `linkModes` 1.3.
- **Changed**
  * The exported package `hipIpInterface` from 1.1 to 1.2.
  * The exported package `hipNetworkStatus` from 1.2 to 1.3.
  * The exported package `hipPhysicalPorts` from 1.0 to 1.1.
  * The exported package `hipCardSettings` from 1.1 to 1.2

1.1

- **Changed**
  * The exported package `hipTsEncoder` is bumped from 1.0 to 1.1.
  * The exported package `hipTsSettings` is bumped from 1.0 to 1.1.
  * The exported package `hipEncStatus` is bumped from 1.3 to 1.4.
  * The exported package `hipEncTransport` is bumped from 1.3 to 1.4.
  * The exported package `hipEncTypes` is bumped from 1.3 to 1.4.
  * The exported package `hipStatus` is bumped from 1.3 to 1.4.
  * The exported package `hipTypes` is bumped from 1.3 to 1.4.

- **Added**
  - Added `nmosStatus` 1.0

2 commonTypes (1.0)

2.1 Type Reference

2.1.1 NmosVersionNumber

Generic version number type for NMOS APIs

enum

V1_2	
V1_3	

3 dpi (1.0)

3.1 Type Reference

3.1.1 Dpi

Configuration of the DPI data

struct

`source`	**DpiSource** Source of the DPI data
`heartbeat`	**bool** Whether heartbeat is on or off
`heartbeatInterval`	**int** Interval between each heartbeat
`ptsOffset`	**int** SCTE-35 PTS adjustment
`index`	**int** SCTE 104 DPI PID index

3.1.2 DpiSettings

Mapping an identifier to the DPI data

struct

`identifier`	**optional int** The identifier for mapping SCTE104 messages to the service.
`dpiData`	**Dpi** Configuration of the DPI data

3.1.3 DpiSource

Source of the DPI data

enum

`ANC`	
`TCP`	

4 dpiStatus (1.0)

4.1 Type Reference

4.1.1 DpiCommand

DPI Command info

struct

`commandType`	**int** Last SCTE command type.
`descriptorIds`	**list** of **int** List of segmentation descriptor IDs.

4.1.2 DpiInput

DPI input status.

struct

`sourceIdentifier`	**optional int** identifies the source of the input. Encoder: ANC is represented by -1, TCP input is represented by its identifier.
`received`	**bool** Received DPI input.
`invalidPackets`	**int** Input contained invalid packets.
`pidIndexes`	**list** of **int**
`timeSinceLastReceived`	**int**
`lastGeneratedCommand`	**optional DpiCommand**

4.1.3 DpiStatus

DPI status.

struct

`heartbeat`	**bool** Specifies whether heartbeat signal is on or off.
`enabled`	**bool** Specifies whether DPI is enabled.
`running`	**bool** Specifies whether DPI encoding is running without errors.
`pidIndex`	**int** SCTE PID index.
`timeSinceLastReceived`	**int** Seconds since last received DPI packet.
`timeSinceLastGenerated`	**int** Seconds since last generated DPI packet.
`lastCommandType`	**int** Type of last received SCTE command.

`discardedPackets`	**int** Number of discarded packets.

5 firewallTypes (1.0)

5.1 Type Reference

5.1.1 IpAddress

string

5.1.2 PaginatedRequest

```
Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request
@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber
```

struct

pageSize	**int**
pageNumber	**int**

5.1.3 PaginationInfo

struct

totalEntries	**int** the total number of entries on the server, before matching

5.1.4 SocketAddress

Tuple with IP address and port

struct

address	**IpAddress** IP address, either IPv4 or IPv6
port	**int**

5.1.5 UUID

string

6 hipCardSettings (1.2)

6.1 Overview

Changelog:

1.2

- **Added**
  - The value nmosConfig is added to cardsettings.

6.2 Command Reference

6.2.1 GetCardSettings

- **message GetCardSettings.Request**
- **message GetCardSettings.Response**

6.2.2 SetCardSettings

CAUTION: Always use the same UUID as the object already existing on the card.

- **message SetCardSettings.Request**
- **message SetCardSettings.Response**

6.3 Type Reference

6.3.1 CardSettings

- Settings for the operational mode of the card.
-
-

struct

`clockMode`	**ClockMode** Type of clock synchronization protocol the card should use •
`nmos`	**optional nmosConfig.NmosConfig** Parameters for accessing NMOS registry and labeling templates

6.3.2 ClockMode

Different types of clock synchronization protocols

enum

PTP	Sets the card to Precision Time Protocol (IEEE 1588)(PTP) time synchronization
LOCAL	Sets the card to Network Time Protocol (NTP) time synchronization or local clock if the card does not get a NTP lock.

6.3.3 GetCardSettings.Request

empty **struct**

6.3.4 GetCardSettings.Response

struct

`data`	**map** from **UUID** to **CardSettings**

6.3.5 SetCardSettings.Request

struct

`data`	**map** from **UUID** to **CardSettings**

6.3.6 SetCardSettings.Response

empty **struct**

7 hipEncStatus (1.6)

7.1 Overview

Changelog:

1.6

- **Added**
  - New fields `pathADelay` and `pathBDelay` in `SeamlessStatus`

1.5

- **Added**
  - New type `DpiTransportStatus`
  - New field `dpiStatus` in `EncoderTransportStatus`
  - `dpiStatus` 1.0.

1.4

- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Added**
  - The Command ClearAllCounters is added.
- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.
  - The fields rtpSsrc and rtpPayloadType on type RTPStatus are changed from being int to being optional(int).
  - The fields packetsMissing and sequenceErrors on type RTPStatus are changed from type int to type bigint.
  - The field rtpStatus on type TransportStatus is no longer optional.

1.1

- **Added**
  - The type SeamlessStatus is added.
  - The field seamlessStatus is added to the type EssenceTransportStatus;
- **Changed**
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.
  - The field timeReference on type NetworkLatencyStatus changes type from hipEncTypes.TimeReference to hipTypes.TimeReference.
- **Removed**

- The imported package `hipEncTypes` 1.0 is removed.

7.2 Command Reference

7.2.1 GetEncoderTransportStatus

- message **GetEncoderTransportStatus.Request**
- message **GetEncoderTransportStatus.Response**

7.2.2 ClearAllCounters

- message **ClearAllCounters.Request**
- message **ClearAllCounters.Response**

7.3 Type Reference

7.3.1 ClearAllCounters.Request

empty **struct**

7.3.2 ClearAllCounters.Response

empty **struct**

7.3.3 DpiTransportStatus

Dpi Transport Status

struct

`dpiInput`	**list** of **dpiStatus.DpiInput** DPI Inputs
`dpiOutput`	**map** from **int** to **dpiStatus.DpiStatus** Maps from DPI PIDs to DPI ouputs

7.3.4 EncoderTransportStatus

Encoder Transport Status

Status is reported for all configured essences.

struct

`videoStatus`	**optional EssenceTransportStatus** Video essence status
`audioStatus`	**map** from **hipTypes.AudioIndex** to **EssenceTransportStatus** Map of audio essence status
`ancStatus`	**optional EssenceTransportStatus** Anc essence status
`dpiStatus`	**optional DpiTransportStatus** DPI Transport Status. Only used by the TS Encoder product

7.3.5 EncoderTransportStatusRequest

struct

`ids`	**optional list** of **UUID**

7.3.6 EncoderTransportStatusResponse

struct

`data`	**map** from **UUID** to **EncoderTransportStatus**

7.3.7 EssenceTransportStatus

Transport Status for an Essence

struct

`pathA`	**optional TransportStatus** Status for port D1 (empty if D1 not configured)
`pathB`	**optional TransportStatus** Status for port D2 (empty if D2 not configured)
`seamlessStatus`	**optional SeamlessStatus** Seamless status (empty if seamless not configured)

7.3.8 GetEncoderTransportStatus.Request

EncoderTransportStatusRequest

7.3.9 GetEncoderTransportStatus.Response

EncoderTransportStatusResponse

7.3.10 NetworkLatencyStatus

Network latency status

struct

`timeReference`	**hipTypes.TimeReference** Time reference protocol
`minDelay`	**float** Minimum timestamp difference (millisec)
`maxDelay`	**float** Maximum timestamp difference (millisec)

7.3.11 RTPStatus

RTP status

struct

bitrate	**float**
packetsMissing	**bigint**
sequenceErrors	**bigint**
rtpPayloadType	**optional int**
rtpSsrc	**optional int**

7.3.12 SeamlessStatus

Seamless status

struct

relativeDelay	**int** Delay between inputs (millisec)
margin	**int** Difference between relative delay and configured delay (millisec)
synchronised	**bool** Inputs are synchronised (bool)
rtpSequenceErrors	**int** Number of RTP sequence errors out of the seamless buffer
pathADelay	**int** Buffer delay for path A (millisec)
pathBDelay	**int** Buffer delay for path B (millisec)

7.3.13 TransportStatus

Transport Status for a Single Path

struct

rtpStatus	**RTPStatus** RTP status
networkLatencyStatus	**optional NetworkLatencyStatus** Network latency status (empty if no bitrate)

8 hipEncTransport (1.5)

8.1 Overview

Changelog:

1.5

- **Changed**
  - Added an entry to InputUdpSeamless which decieds which redundancy mode to use.
  - Removed identical entry

1.4

- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.
  - `InputUdpSettings` sources field is changed to an optional string named source, to reflect the fact we only support a single source for the IGMPv3 SSM.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.

1.1

- **Changed**
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.

8.2 Type Reference

8.2.1 InputUdpSeamless

Set the input in seamless transport mode (SMTPE-2022-7)

The two sources, a and b, must be setup at different interfaces

struct

a	**InputUdpSettings** UDP settings for the A pipe
b	**InputUdpSettings** UDP settings for the B pipe
`preferred`	**InputUdpSeamless.preferred** Preferred pipe
`inputBufferSize`	**int** Buffer for reconstructing missing packets on the input (1-400ms) [default 20ms]
`redundancyMode`	**RedundancyMode** Type seamless strategy: 2022-7 or A/B failover

8.2.2 InputUdpSeamless.preferred

enum

A
B
FLOATING

8.2.3 InputUdpSettings

UDP input settings

If destination is a multicast address IGMP join will be performed. A destination must be unique for a given interface.

struct

interfaceId	**UUID** Interface used
destination	**hipTypes.SocketAddress** Unicast or multicast address
source	**optional string** Optional IGMPv3 source address

8.2.4 RedundancyMode

Redundancy input settings

enum

SMPTE_2022_7	Multiple sources in seamless mode (SMPTE-2022-7)
AB_FAILOVER	A/B switching on error

8.2.5 SourceSettings

Source Settings of UDP input

variant

seamless	**InputUdpSeamless** Multiple sources in seamless mode (SMPTE-2022-7)
single	**InputUdpSettings** Single source

9 hipEncTypes (1.4)

9.1 Overview

Changelog:

1.4

- **Changed**
- The imported package `hipTypes` is bumped from 1.3 to 1.4.

1.3

- **Changed**
- The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Added**
  - The imported package `hipTypes` 1.2 is added.
- **Changed**
  - The type of the field format on type VideoPolicing are changed from VideoFormat to hipTypes.VideoFormat.
- **Removed**
  - The types ResolutionScan, FrameRate and VideoFormat are removed.

1.1

```
- **Removed**
- The type TimeReference is removed.
```

9.2 Type Reference

9.2.1 AncPolicing

Ancillary Policing

Policing on the ancillary essence.

struct

`maxBitrate`	**optional float** (Optional) Maximum policed bitrate of the ancillary essence [Mbps]

9.2.2 AudioPolicing

Audio Policing

Police on the Audio essences.

struct

optional float

`maxBitrate`	(Optional) Maximum policed bitrate of the audio essence [Mbps]

9.2.3 OptimalizationTarget

Optimalization Target

Target of encoding optimalization

enum

`VISUAL`	Optimize for optimal visuals
`PSNR`	Optimize for maximized Peak Signal To Noise Ratio

9.2.4 VideoPolicing

Video Policing

Policing on the video essence.

struct

`maxBitrate`	**optional float** (Optional) Maximum policed bitrate of the video essence [Mbps]
`format`	**optional hipTypes.VideoFormat** (Optional) Allowed format of input video

10 hipIpInterface (1.2)

10.1 Overview

Changelog:

1.2

- **Changed**
  - The imported package `hipPhysicalPorts` is bumped from 1.0 to 1.1.

1.1

- **Added**
  - The `lldp` field is added to the `IpInterface` object.
  - The `GetLldpNeighbors` RPC is added

10.2 Command Reference

10.2.1 GetIpInterfaces

- **message GetIpInterfaces.Request**
- **message GetIpInterfaces.Response**

10.2.2 SetIpInterfaces

- **message SetIpInterfaces.Request**
- **message SetIpInterfaces.Response**

10.2.3 DeleteIpInterfaces

- **message DeleteIpInterfaces.Request**
- **message DeleteIpInterfaces.Response**

10.2.4 GetLldpNeighbors

Get LLDP Neighbors.

Returns data about the LLDP neighbor seen per interface. To see data for a specific port, LLDP must be enabled for that Interface.

- **message GetLldpNeighbors.Request**
- **message GetLldpNeighbors.Response**

10.3 Type Reference

10.3.1 DeleteIpInterfaces.Request

struct

`ids`	**list** of **UUID**

10.3.2 DeleteIpInterfaces.Response

empty **struct**

10.3.3 GetIpInterfaces.Request

empty **struct**

10.3.4 GetIpInterfaces.Response

struct

`data`	**map** from **UUID** to **IpInterface**

10.3.5 GetLldpNeighbors.Request

GetLldpNeighborsRequest

10.3.6 GetLldpNeighbors.Response

GetLldpNeighborsResponse

10.3.7 GetLldpNeighborsRequest

Get LLDP Neighbors Request

struct

`interface`	**optional list** of **UUID** List of UUIDs uniquely identifying IpInterfaces. Only LLDP Neighbors from the specified IpInterfaces are returned. If no value is set, the request will return neighbors from all ports that have LLDP enabled.
`details`	**bool** If false only chassisID, mgmtIP, and portID will be returned for each LLDP Neighbor. If true all available information is returned.

10.3.8 GetLldpNeighborsResponse

struct

`data`	**map** from **UUID** to **optional lldp.LldpNeighbor**

10.3.9 IpInterface

IP Interface

struct

`label`	**string** User label

physicalPortName	**physicalIpPort.PortName** The physical port used by the ip interface
enabled	**bool** Enable/disable the IP interface
vlan	**int** Vlan number. vlan 0 indicates no vlan in use.
quad	**optional QuadPortName** Interface number when physical port is set to quad mode (4x10G). Is optional.
ipv4	**optional IpInterfaceConfig** IPv4 interface configuration. Is optional.
ipv6	**optional IpInterfaceConfig** IPv4 interface configuration. Is optional.
lldp	**optional lldp.LldpMode** The LldpMode to set for the port. Leaving this empty indicates LLDP off.

10.3.10 IpInterfaceConfig

IP Interface configuration

struct

ipAddress	**string** IP address
gateway	**string** Gateway address of the interface
netmask	**int** Netmask of the interface in CIDR subnet mask notation e.g /24 = 255.255.255.0

10.3.11 QuadPortName

enum

Q1	
Q2	
Q3	
Q4	

10.3.12 SetIpInterfaces.Request

struct

data	**map** from **UUID** to **IpInterface**

10.3.13 SetIpInterfaces.Response

empty **struct**

11 hipNetworkStatus (1.3)

11.1 Overview

Changelog:

1.3

- Changed
  - The imported package `hipPhysicalPorts` is bumped from 1.0 to 1.1.

1.2

- Changed
  - The imported package `sfpStatus` is bumped from 1.3 to 1.4.

1.1

- Added
  - The imported package `sfpStatus` 1.3 is added.
  - The command GetSfpStatus is added.

11.2 Command Reference

11.2.1 GetNetworkStatus

- message GetNetworkStatus.Request
- message GetNetworkStatus.Response

11.2.2 GetSfpStatus

- message GetSfpStatus.Request
- message GetSfpStatus.Response

11.3 Type Reference

11.3.1 GetNetworkStatus.Request

NetworkStatusRequest

11.3.2 GetNetworkStatus.Response

NetworkStatusResponse

11.3.3 GetSfpStatus.Request

GetSfpStatusRequest

11.3.4 GetSfpStatus.Response

GetSfpStatusResponse

11.3.5 GetSfpStatusRequest

Get SFP Status Request

struct

physicalports	**optional list** of **UUID** List of UUIDs uniquely identifying PhysicalPorts. Only SFP Status from the specified PhysicalPorts are returned. If no value is set, the request will return SFP Status from all ports.

11.3.6 GetSfpStatusResponse

struct

data	**map** from **UUID** to **sfpStatus.SfpStatus**

11.3.7 LinkSpeed

LinkSpeed

enum

NO_LINK	No linkspeed
LINK_10_MBPS	10 Mbps
LINK_100_MBPS	100 Mbps
LINK_1_GBPS	1 Gbps
LINK_10_GBPS	10 Gbps
LINK_25_GBPS	25 Gbps
LINK_40_GBPS	40 Gbps

11.3.8 NetworkStatus

NetworkStatus

struct

macAddress	**string** Interface MAC address.
linkSpeed	**LinkSpeed** Linkspeed on interface.
rx	**optional int** Received bitrate on interface (kbps). Only applicable for the IP JPEG XS Encoder.
tx	**optional int** Transmitted bitrate on interface (kbps). Only applicable for the IP JPEG XS Decoder.

11.3.9 NetworkStatusRequest

NetworkStatusRequest

struct

`ids`	**optional list** of **UUID** Optional list of interface uuids

11.3.10 NetworkStatusResponse

NetworkStatusResponse

struct

`data`	**map** from **UUID** to **NetworkStatus** Map of networkstatus with interface uuid as key

12 hipPhysicalPorts (1.1)

12.1 Overview

Changelog:

1.1

- **Changed**
  - The `portMode` field in `PhysicalPort` was changed to `linkMode` and is using the `IpLinkMode` type from `linkModes` 1.3.
- **Added**
  - A `fecMode` field was added to the `PhysicalPort` object using the `IpFecMode` type from `linkModes` 1.3.

12.2 Command Reference

12.2.1 GetPhysicalPorts

- message **GetPhysicalPorts.Request**
- message **GetPhysicalPorts.Response**

12.2.2 SetPhysicalPorts

- message **SetPhysicalPorts.Request**
- message **SetPhysicalPorts.Response**

12.3 Type Reference

12.3.1 GetPhysicalPorts.Request

empty **struct**

12.3.2 GetPhysicalPorts.Response

GetPhysicalPortsResponse

12.3.3 GetPhysicalPortsResponse

struct

`data`	**map** from **UUID** to **PhysicalPort**

12.3.4 PhysicalPort

Physical Port

struct

`name`	**physicalIpPort.PortName** Name of the physical port

`label`	**string** User label
`linkMode`	**linkModes.IpLinkMode** Port mode determines link speed and quad mode
`fecMode`	**optional linkModes.IpFecMode** Optional FEC mode
`enabled`	**bool** Enable / disable the physical port

12.3.5 SetPhysicalPorts.Request

SetPhysicalPortsRequest

12.3.6 SetPhysicalPorts.Response

empty **struct**

12.3.7 SetPhysicalPortsRequest

struct

`data`	**map** from **UUID** to **PhysicalPort**

13 hipStatus (1.5)

13.1 Overview

Changelog:

1.5

- **Changed**
  - The field allocatedCoreType was added to type VideoStatus.

1.4

- **Changed**
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.

1.3

- **Changed**
  - The imported package `hipTypes` is bumped from 1.2 to 1.3.

1.2

- **Changed**
  - The imported package `hipTypes` is bumped from 1.1 to 1.2.
  - The field format on type VideoStatus is now optional.

1.1

- **Added**
  - The command GetThumbnailPath is added.
  - The field lineNumber is added to type DataIdentifier.
- **Changed**
  - The imported package `hipTypes` is bumped from 1.0 to 1.1.

13.2 Command Reference

13.2.1 GetEssenceStatus

Get Encoder essence status

- **message GetEssenceStatus.Request**
- **message GetEssenceStatus.Response**

13.2.2 GetThumbnailPath

Get Video thumbnail path

- **message GetThumbnailPath.Request**
- **message GetThumbnailPath.Response**

13.3 Type Reference

13.3.1 AncStatus

Ancillary Status

struct

`dataIdentifiers`	**list** of **DataIdentifier** List of data identifiers

13.3.2 AudioStatus

Audio Status

struct

`totalChannels`	**int** Total number of of Audio Channels
`standard`	**optional** **st2110Types.St2110Standard** St2110 audio standard
`packetTime`	**hipTypes.St2110AudioPacketTime** Packet time

13.3.3 DataIdentifier

Data Identifiers

The Data Identifier (DID) and Secondary Data Identifier (SDID) words which together indicate a type of ancillary data.

struct

`did`	**int** Data Identifier word
`sdid`	**int** Secondary Data Identifier word
`lineNumber`	**int** Line number of the received DID/SDID pair

13.3.4 EssenceStatus

Essence Status

Status is reported for all configured essences that have bitrate.

struct

`videoStatus`	**optional** **VideoStatus** Video status
`audioStatus`	**map** from **hipTypes.AudioIndex** to **AudioStatus** Map of audio statuses
`ancStatus`	**optional** **AncStatus** Ancillary status

13.3.5 EssenceStatusRequest

Essence Status Request

struct

`ids`	**optional list** of **UUID** Optional list of Encoder/Decoder UUIDs to request status for. If ids are empty, status for all UUIDs present will be returned.

13.3.6 EssenceStatusResponse

Essence Status Response

struct

`data`	**map** from **UUID** to **EssenceStatus** Map from Encoder/Decoder UUID to the related essence status structure.

13.3.7 EssenceThumbnailPathRequest

Essence Thumbnail Path Request

struct

`ids`	**optional list** of **UUID** Optional list of Encoder/Decoder UUIDs to request thumbnail path for. If ids are empty, status for all UUIDs present will be returned.

13.3.8 EssenceThumbnailPathResponse

Essence Thumbnail Path Response

struct

`data`	**map** from **UUID** to **string** Map from Encoder/Decoder UUID to the related essence thumbnail path.

13.3.9 GetEssenceStatus.Request

EssenceStatusRequest

13.3.10 GetEssenceStatus.Response

EssenceStatusResponse

13.3.11 GetThumbnailPath.Request

EssenceThumbnailPathRequest

13.3.12 GetThumbnailPath.Response

EssenceThumbnailPathResponse

13.3.13 VideoStatus

- Video Status
-
-

struct

`compressed`	**bool** True if compressed, and false if raw •
`format`	**optional hipTypes.VideoFormat** Video Resolution and Frame Rate •
`allocatedCoreType`	**optional hipTypes.CodingCore** Core type allocated internally for decoding, is null on failed core allocation. On core allocation, UHD cores will be automatically utilized by HD/SD configurations if all 4 HD cores are already in use.

14 hipTestGenerator (1.1)

14.1 Type Reference

14.1.1 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

enum

BLACK	
TRANSPARENT	

14.1.2 Color

A color defined in terms of RGB.

struct

red	int
green	int
blue	int

14.1.3 MovingBox

Specify the properties of a moving box to be used in a test image generator.

struct

enabled	bool
color	optional Color

14.1.4 TestPattern

Pattern to display through generator

enum

BLACK_PICTURE	Display black picture
COLOR_BARS	Display color bars

14.1.5 TextOverlay

Specify the properties of a text overlay to put on top of a test image.

Uses the label of the De/Encoder object as text to display. Text will be truncated after 36 characters

struct

enabled	bool
position	TextPosition
background	BitmapBackgroundProperty

14.1.6 TextPosition

Specify the vertical placement for the text.

enum

TOP
MIDDLE
BOTTOM

14.1.7 VideoTestGenerator

Defines how to generate a test image for a specific port.

struct

forceGenerator	**bool**
pattern	**TestPattern**
textOverlay	**TextOverlay**
movingBox	**MovingBox**
audio	**bool**

15 hipTsEncoder (1.7)

15.1 Overview

Changelog:

1.7

```
  - **Changed**
  - `videoUsabilityInfo` from 1.1 to 1.2
```

1.6

```
  - **Changed**
  - `hipTestGenerator` from 1.0 to 1.1
```

1.5

```
  - **Changed**
  - `hipEncTransport` from 1.4 to 1.5
```

1.4

```
- **Changed**
  - The imported package `hipEncTransport` is bumped from 1.3 to 1.4.
  - The imported package `videoUsabilityInfo` is bumped from 1.0 to 1.1.
```

1.3

```
- **Changed**
  - The imported package `hipTsSettings` is bumped from 1.1 to 1.2.
```

1.2

- **Changed**
  - Changed vuiParams on EncoderVideoEssence to colorProfile

1.1

```
- **Added**
  - Added `vuiParams` to `EncoderVideoEssence`
  - Imported package `videoUsabilityInfo` version 1.0
- **Changed**
  - The imported package `hipTsSettings` is bumped from 1.0 to 1.1.
  - The imported package `hipEncTransport` is bumped from 1.3 to 1.4.
  - The imported package `hipTypes` is bumped from 1.3 to 1.4.
  - `numChannels` is moved from `EncoderAudioEssence` into  `tsSettings` in `Encoder`
```

15.2 Command Reference

15.2.1 GetEncoders

Get Encoder objects from database.

- **message GetEncoders.Request**
- **message GetEncoders.Response**

15.2.2 SetEncoders

Update Encoder objects in database.

- message SetEncoders.Request
- message SetEncoders.Response

15.2.3 DeleteEncoders

Delete Encoder objects from database.

- message DeleteEncoders.Request
- message DeleteEncoders.Response

15.3 Type Reference

15.3.1 ColorProfile

variant

colorType	ColorType
vuiParams	videoUsabilityInfo.VuiParameters

15.3.2 ColorType

enum

HLG_10
PQ_10
SDR
SDR_WCG

15.3.3 DeleteEncoders.Request

struct

ids	**list** of **UUID**

15.3.4 DeleteEncoders.Response

empty **struct**

15.3.5 Encoder

Encoder Config

Encoder configuration

struct

enable	**bool** Enable / disable encoder
	string

label	User label
source	**EncoderSource2110** ST2110 source struct
onSignalLoss	**hipTypes.SignalLossAction** Action on signal loss
tsSettings	**hipTsSettings.TsSettings** TS encapsulation settings

15.3.6 EncoderAncEssence

Encoder TS Anc Essence

Specification for TS ancillary essence source

struct

settings	**EncoderEssenceSettings** Essence settings
policing	**hipEncTypes.AncPolicing** Ancillary policing rules

15.3.7 EncoderAudioEssence

Encoder TS Audio Essence

Specification for TS audio essence source

struct

settings	**EncoderEssenceSettings** Essence settings
st2110Standard	**st2110Types.St2110Standard** ST2110 audio standard
policing	**hipEncTypes.AudioPolicing** Essence policing rules

15.3.8 EncoderEssenceSettings

Encoder ST2110 Essence Settings

Settings for ST2110 essence source

struct

enable	**bool** Enable essence
label	**string** User label of essence
source	**hipEncTransport.SourceSettings** Transport settings

15.3.9 EncoderSource2110

Encoder TS Source

Specification of TS source

struct

`videoEssence`	**EncoderVideoEssence** Video essence of source
`audioEssences`	**map** from **hipTypes.AudioIndex** to **EncoderAudioEssence** Map over audio essences
`ancEssence`	**optional** **EncoderAncEssence** Optional ancillary essence

15.3.10 EncoderVideoEssence

Encoder TS Video Essence

Specification for TS video essence source

struct

`settings`	**EncoderEssenceSettings** Essence settings
`policing`	**hipEncTypes.VideoPolicing** Essence policing rules
`testGenerator`	**hipTestGenerator.VideoTestGenerator** Video TestGenerator settings
`encoding`	**JpegXsEncoding** Encoding settings
`colorProfile`	**ColorProfile** Predefined Color Type or manual Vui parameters

15.3.11 GetEncoders.Request

empty **struct**

15.3.12 GetEncoders.Response

struct

`data`	**map** from **UUID** to **Encoder**

15.3.13 JpegXsEncoding

Jpeg Xs Encoding Settings

struct

`compressedBitrate`	**float** Target bitrate [Mbps]
	hipEncTypes.OptimalizationTarget

`optimizationTarget`	Optimization Target
`coreType`	**hipTypes.CodingCore** Type of encoding core

15.3.14 SetEncoders.Request

struct

`data`	**map** from **UUID** to **Encoder**

15.3.15 SetEncoders.Response

empty **struct**

16 hipTsSettings (1.2)

16.1 Overview

Changelog:

1.2

```
- **Changed**
  - Added `dpiSettings` to `TsSettings`
```

1.1

```
- **Changed**
  - Added `numChannels` to `AudioSettings`
  - Added `startChannel` to `AudioSettings`
  - Added `enable` to `AudioSettings`
```

16.2 Type Reference

16.2.1 AudioSettings

Audio Settings Settings for Audio Pid

struct

`enable`	**bool** Play out pid in the resulting ts
`essence`	**hipTypes.AudioIndex** Essence included in PID
`numChannels`	**hipTypes.NumberOfChannels** Number of channels in PID
`startChannel`	**hipTypes.StartChannel** First channel in essence to put into PID

16.2.2 ServiceSettings

Service Settings

Service level configuration

struct

`serviceName`	**string** Name of the Service
`serviceId`	**int** Service ID of the Service
`pcrPid`	**int** PCR PID for service
`pcrInterval`	**int** Interval in which PCR is signalled repeatedly (in milliseconds)
`pmtPid`	**int** PMT PID for service

16.2.3 TsSettings

TS settings Settings for generated TS

struct

`serviceSettings`	**ServiceSettings** Service Configuration
`videoPid`	**int** Video PID
`audioSettings`	**map** from **int** to **AudioSettings** Maps Audio PID to AudioSettings
`ancPid`	**optional int** Anc PID (optional)
`dpiSettings`	**map** from **int** to **dpi.DpiSettings** Maps DPI PID to DpiSettings

17 hipTypes (1.4)

17.1 Overview

Changelog:

1.4

- **Added**
  - The enum StartChannel

1.3

- **Added**
  - The value R2160P to ResolutionScan

1.2

- **Removed**
  - The values BLACK_PICTURE, MOVING_BOX and COLOR_BARS are removed from type Signal-LossActionVideo.

1.1

- **Added**
  - The type TimeReference is added.

17.2 Type Reference

17.2.1 AudioIndex

Audio Index

enum

AUDIO_1	Audio index 1
AUDIO_2	Audio index 2
AUDIO_3	Audio index 3
AUDIO_4	Audio index 4
AUDIO_5	Audio index 5
AUDIO_6	Audio index 6
AUDIO_7	Audio index 7
AUDIO_8	Audio index 8

17.2.2 CodingCore

Coding Core Type

Type of Coding Core to use.

enum

UHD	Use Core that supports UHD/HD/SD
HD	Use Core that supports HD/SD

17.2.3 FrameRate

Frame Rate

Frame rate of the source.

enum

F24	24 fps
F23_98	23.98 fps
F25	25 fps
F29_97	29.97 fps
F30	30 fps
F47_95	47.95 fps
F48	48 fps
F50	50 fps
F59_94	59.94 fps
F60	60 fps

17.2.4 NumberOfChannels

Number of Channels

enum

CHANNELS_2	2 Mono Channels
CHANNELS_4	4 Mono Channels
CHANNELS_6	6 Mono Channels
CHANNELS_8	8 Mono Channels

17.2.5 Order

Order to put audio channel in.

enum

FIRST	Put audio channels first
SECOND	Put audio channels second
THIRD	Put audio channels third
FOURTH	Put audio channels fourth
FIFTH	Put audio channels fifth
SIXTH	Put audio channels sixth
SEVENTH	Put audio channels seventh
EIGHTH	Put audio channels eighth

17.2.6 ResolutionScan

Resolution and Scan

Resolution and scan mode of the source.

enum

R2160P	2096x2160 progressive

R2KP	2048x1080 progressive
R1080P	1920x1080 progressive
R1080I	1920x1080 interlaced
R720P	1280x720 progressive
R576I	720x576 interlaced
R480I	720x480 interlaced

17.2.7 SignalLossAction

Signal Loss Action

Action taken on loss of bitrate on essences.

struct

videoAction	**SignalLossActionVideo** Action on video essence
audioAction	**SignalLossActionAudio** Action on audi essences
ancAction	**SignalLossActionAnc** Action on ancillary essence

17.2.8 SignalLossActionAnc

Signal Loss Action on ancillary essence

Action taken on loss of bitrate on the ancillary essence.

enum

MUTE	Mutes ancillary essence
TEST_GEN	Generates test ancillary essence

17.2.9 SignalLossActionAudio

Signal Loss Action on audio essences

Action taken on loss of bitrate on the audio essences.

enum

MUTE	Mutes audio essence
TEST_GEN	Generates test audio

17.2.10 SignalLossActionVideo

Signal Loss Action on video essence

Action taken on loss of bitrate on the video essence.

enum

MUTE	Mutes video essence
TEST_GEN	Display picture from test generator

17.2.11 SocketAddress

Socket address

Tuple with IP address and port

struct

address	**string** IP address, either IPv4 or IPv6
port	**int** Port number

17.2.12 St2110AudioPacketTime

ST2110 Audio packet time

enum

N_A	No audio packet time
US83_3	83,3 us packet time
US125	125 us packet time
MS1	1 ms packet time

17.2.13 StartChannel

Channel to start on, starts at 1 up to 15 (Only even numbered channel pairs)

enum

CHANNEL_1	Starting at Channel 1
CHANNEL_3	Starting at Channel 3
CHANNEL_5	Starting at Channel 5
CHANNEL_7	Starting at Channel 7
CHANNEL_9	Starting at Channel 9
CHANNEL_11	Starting at Channel 11
CHANNEL_13	Starting at Channel 13
CHANNEL_15	Starting at Channel 15

17.2.14 TimeReference

Time Reference

Protocol used for time reference

enum

NTP	NTP (Network Time Protocol) time source
PTP	PTP (Precision Time Protocol) time source

17.2.15 VideoFormat

Format of the video essence.

struct

`resolutionScan`	**ResolutionScan** Resolution and scan mode of the source
`frameRate`	**FrameRate** Framerate of the source

18 hipValidate (1.0)

18.1 Type Reference

18.1.1 FloatRange

Float Range

struct

`min`	**float** Minimum value of the range
`max`	**float** Maximum value of the range

18.1.2 IntRange

Integer Range

struct

`min`	**int** Minimum value of the range
`max`	**int** Maximum value of the range

18.1.3 Result

Validation result

struct

`result`	**bool** True if config is valid and false otherwise.
`msg`	**optional string** An error message, which may be given if the result is invalid.

19 linkModes (1.3)

19.1 Type Reference

19.1.1 IpFecMode

enum

NO_FEC
RS_FEC
FC_FEC

19.1.2 IpLinkMode

enum

IP_10G
IP_25G_COPPER
IP_25G_OPTICAL
IP_40G
IP_4x10G

19.1.3 IpLinkSpeed

enum

IP_10G
IP_25G
IP_40G
IP_4x10G
NO_LINK

20 lldp (1.1)

20.1 Overview

```
# Changelog

## 1.1
- **Changed**
    - Extracted the structure inside of the `LldpNeighbor` object definition into a new type cal
```

20.2 Type Reference

20.2.1 LldpMedInventory

LLDP-MED Inventory. Appear specific implementation in parentheses.

struct

`hwRev`	**string** Hardware Revision
`swRev`	**string** Software Revision (Software Version)
`fwRev`	**string** Firmware Revision (FPGA Version)
`serial`	**string** Serial Number
`manufacturer`	**string** Manufacturer ("Appear")
`model`	**string** Model ("X10" or "X20")
`assetId`	**string** Asset ID (Hardware ID)

20.2.2 LldpMode

Choose either private or public configuration of LLDP frames.

enum

`PUBLIC`	Only mandatory TLVs and port detail TLVs.
`PRIVATE`	LLDP-MED inventory in addition to the PUBLIC TLVs.

20.2.3 LldpNeighbor

LldpNeighborStruct

20.2.4 LldpNeighborStruct

The optional variables are only present if GetLldpNeighborRequest.details is set to true in the rpc request. The following descriptions are the Appear specific implementations of each LLDP TLV. If neighbors from other manufacturers are detected, TLV values might differ. The LldpMode is specified in parentheses for variables that have different implementations based on the LldpMode set in the neighbor.

struct

`chassisId`	**string** MAC address of CTRL on MMI 1, fallback to CTRL on MMI 2 (PRIVATE) or “N/A” (PUBLIC).
`mgmtIp`	**list** of **string** IPv4 address of CTRL on MMI 1 and/or CTRL on MMI 2. Also IPv6 if enabled on the neighbor.
`sysDescr`	**optional string** Desktop Heading (PRIVATE) or “Unknown” (PUBLIC)
`sysName`	**optional string** Hostname (PRIVATE) or Interface.Port (PUBLIC)
`portId`	**string** MAC address of local port
`portDescr`	**optional string** Slot, Interface: Port-label
`ttl`	**optional string** Time-To-Live, how long an LLDP packet received from that neighbor should be considered valid.
`lldpMedInventory`	**optional LldpMedInventory** LLDP-MED Inventory

21 networkManagerProtocols (1.1)

21.1 Overview

Changelog:

1.1

- **Changed**
  - The imported package `hipIpInterface` is bumped from 1.1 to 1.2.

1.0

- copied from firewall networkManagerProtocol.1.7

22 nmosConfig (1.0)

22.1 Type Reference

22.1.1 NmosConfig

Custom labels are enabled if the optional is set.

struct

`registry`	nmosRegistryConfig.NmosRegistryConfigStruct
`labelTemplates`	optional nmosLabelConfig.NmosLabelConfig

23 nmosLabelConfig (1.0)

23.1 Overview

Changelog

1.0

NMOS labelling

NMOS labelling API allows operator to define templates based on auto text options. This templates can be defined for for node, device, source, flow, sender and receiver.

Version 1.0 allows six auto text options.

What are the auto text options?

- ${SLOT} - The slot number of the card in use.
- ${DEV_ID} - The Device number. **Not supported for Node templates**.
- ${ESS_GR} - The label of the group where the essences belong to. **Not supported for Node templates**.
- ${ESS} - The label of the Essence in question. **Not supported for Node and Device templates**.
- ${ESS_T} - The Essence type: Video, Audio, Ancillary. This auto text option is substituted by what is defined in `EssenceTypeStrings` for each of the essences type. **Not supported for Node and Device templates**.
- ${ESS_T_INDEX} - The number of the essence belonging to one of the essence type - video, audio, ancillary. **Not supported for Node and Device templates**.

How to use the NMOS Labelling.

- NMOS templates are set in the fields part of `ResourceLabelTemplates`. Unused templates should be set with an empty string.
- Each field in `ResourceLabelTemplates` can have as many auto text options as the operator wants, as long as they are supported for the respective template.
- `EssenceTypeStrings` strings will substitute the auto text option ${ESS_T} with the defined string for the respective Essence type.

Example of strings to use for the fields in `ResourceLabelTemplates` and `EssenceTypeStrings`

- `nodeTemplate` - ECx210 Decoder ST2110 Sender Slot ${SLOT}
- `deviceTemplate` - ECx210 Decoder Sender Slot ${SLOT} ID ${DEV_ID}
- `sourceTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Source: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `flowTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Flow: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `senderTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Sender: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `receiverTemplate` - ECx210 Slot ${SLOT} ID ${DEV_ID}, ${ESS_T} Receiver: ${ESS}, Group: ${ESS_GR}, Essence Number: ${ESS_T_INDEX}
- `video` - Video
- `audio` - Audio
- `ancillary` - Anc

23.2 Type Reference

23.2.1 EssenceTypeStrings

Labels for each essence type to use when generating custom NMOS labels.

struct

`video`	**string** Video essence label
`audio`	**string** Audio essence label
`ancillary`	**string** Ancillary essence label

23.2.2 NmosLabelConfig

Custom NMOS labels configuration.

struct

`resourceTemplates`	**ResourceLabelTemplates** Templates for each NMOS resource
`essenceStrings`	**EssenceTypeStrings** Labels to use for each essence type

23.2.3 ResourceLabelTemplates

NMOS label templates for each NMOS resource.

struct

`nodeTemplate`	**string** Template string for the node label
`deviceTemplate`	**string** Template string for the device label
`sourceTemplate`	**string** Template string for the source label
`flowTemplate`	**string** Template string for the flow label
`senderTemplate`	**string** Template string for the sender label
`receiverTemplate`	**string** Template string for the receiver label

24 nmosRegistryConfig (1.0)

24.1 Type Reference

24.1.1 DnsConfig

Method to use for discovery of registry server

variant

`mDns`	**DnsConfig.mDns** Use mDNS to reach the registry server
`dnsSd`	**DnsSdConfig** Configuration parameters to use DNS-SD for discovery of registry service

24.1.2 DnsConfig.mDns

empty **struct**

24.1.3 DnsSdConfig

Configuration parameters for DNS-SD

struct

`dnsServer`	**firewallTypes.IpAddress** IPv4 address of DNS-SD server
`search`	**string** Search domain name

24.1.4 NmosRegistryConfigStruct

Configuration parameters for reaching the NMOS registry

struct

`registryConfig`	**RegistryManualConfig** Configuration parameters to use in case of failure in mDNS or DNS-SD (secondary/fallback method)
`dnsConfig`	**DnsConfig** Configuration parameters for mDNS or DNS-SD (primary method)

24.1.5 RegistryManualConfig

Registry information to be used in case registry server can not be reached by mDNS or DNS-SD.

struct

`fallbackEnable`	**bool** Enable use of given information as a fallback
	firewallTypes.IpAddress

`registryIpAddr`	IPv4 address of NMOS registry
`registryPort`	**int** Port number of NMOS registry
`registryVersion`	**commonTypes.NmosVersionNumber** Version of registry API

25 nmosRegistryStatus (1.0)

25.1 Type Reference

25.1.1 NmosRegistryStatus

NMOS registry status

struct

`ipAddr`	**optional firewallTypes.SocketAddress** Currently used IP address and port number of the registry server. This value may have been obtained from DNS-SD, mDNS, or a manual IP entry.
`registered`	**bool** True if there is an active connection to the registry server.

26 nmosStatus (1.1)

26.1 Overview

Changelog:

1.1

```
- **Added**
    - The field 'nodeApi' was added to 'NmosStatus'.
- **Changed**
    - The type NmosStatus was moved to the file 'nmosStatusTypes'.
```

26.2 Command Reference

26.2.1 GetNmosStatus

RPC for requesting NMOS status

- **message GetNmosStatus.Request**
- **message GetNmosStatus.Response**

26.3 Type Reference

26.3.1 GetNmosStatus.Request

empty **struct**

26.3.2 GetNmosStatus.Response

nmosStatusTypes.NmosStatus

27 nmosStatusTypes (1.0)

27.1 Type Reference

27.1.1 NmosStatus

NmosStatus Status of the NMOS service

struct

`nmosRegistryStatus`	**nmosRegistryStatus.NmosRegistryStatus** Status of the NMOS registry connection
`nodeApi`	**map** from **physicalIpPort.PortName** to **string** Map from port to the endpoint for NMOS Node API. Ex: PortName.D1 -> "10.20.10.34:8001", PortName.D2_2 -> "10.20.11.34:8001"

28 physicalIpPort (1.0)

28.1 Type Reference

28.1.1 PortName

enum

CTRL	
D1	
D2	
D3	
D4	
D1_1	
D1_2	
D1_3	
D1_4	
D2_1	
D2_2	
D2_3	
D2_4	
D3_1	
D3_2	
D3_3	
D3_4	
D4_1	
D4_2	
D4_3	
D4_4	

29 sfpConstants (1.3)

29.1 Type Reference

29.1.1 Identifier

Physical Device Identifier Values. (SFF 8472 Rev12.3 spec Table 5-1)

enum

UNKNOWN	Unknown or unspecified
GBIC	GBIC
SOLDERED	Module soldered to motherboard (ex. SFF)
SFP_SFPP	SFP or SFP+
QSFPP	QSFP+
QSFP28	QSFP28
NOT_USED	Not used by SFF 8472 Rev12.2.1. These values are maintained in the Transceiver Management section of SFF-8024.
VENDOR	Vendor specific

29.1.2 Transceiver

Transceiver Compliance Codes. (SFF 8472 Rev12.3 spec Table 5-3)

enum

ER_10G	10G Base-ER
LRM_10G	10G Base-LRM
LR_10G	10G Base-LR
SR_10G	10G Base-SR
ACTIVE_CABLE_40G	40G Active cable
LR4_40G	40G Base-LR4
SR4_40G	40G Base-SR4
CR4_40G	40G Base-CR4
PX	BASE-PX
BX10	BASE-BX10
FX_100	100BASE-FX
LX_100	100BASE-LX/LX10
T_1000	1000BASE-T
CX_1000	1000BASE-CX
LX_1000	1000BASE-LX
SX_1000	1000BASE-SX
ELECTRICAL_INTER_ENCLOSURE	Electrical inter enclosure
LONGWAVE_LASER_LL	Longwave laser LL
MEDIUM	Medium distance
LONG_DISTANCE	Long distance
INTERMEDIATE_DISATANCE	Intermediate distance
SHORT_DISTANCE	Short distance
VERY_LONG_DISTANCE	Very long distance
ACTIVE_CABLE	Active Cable
PASSIVE_CABLE	Passive Cable

LONGWAVE_LASER_LC	Longwave laser LC
SHORTWAVE_LASER_w_OFC	Shortwave laser w OFC
SHORTWAVE_LASER_wo_OFC	Shortwave laser w/o OFC
ELECTRICAL_INTRA_ENCLOSURE	Electrical intra enclosure
SINGLE_MODE	Singel mode
MULTIMODE_50um	Multimode 50um
MULTIMODE_50m	Multimode 50m
MULTIMODE_625m	Multimode 62.6m
VIDEO_COAX	Video coax
MINIATURE_COAX	Miniature coax
SHIELDED_TWISTED_PAIR	Shielded twisted pair
TWIN_AXIAL_PAIR	Twin axial pair
M_BYTES_SEC_100	100M bytes per sec
M_BYTES_SEC_200	200M bytes per sec
M_BYTES_SEC_400	400M bytes per sec
M_BYTES_SEC_1600	1600M bytes per sec
M_BYTES_SEC_800	800M bytes per sec
M_BYTES_SEC_1200	1200M bytes per sec Extended Specification Compliance. (SFF-8024 Rev 4.8 spec Table 4-4)
AOC_100G_AOC_25GAUI_C2M_BER_5_	100G AOC (Active Optical Cable) or 25GAUI C2M AOC. Providing a worst BER of $5 * 10^{-5}$
SR4_100G_BASE_SR_25G_BASE	100GBASE-SR4 or 25GBASE-SR
LR4_100G_BASE_LR_25G_BASE	100GBASE-LR4 or 25GBASE-LR
ER4_100G_BASE_ER_25G_BASE	100GBASE-ER4 or 25GBASE-ER
SR10_100G_BASE	100GBASE-SR10
CWDM4_100G	100G CWDM4
PSM4_PARALLEL_SMF_100G	100G PSM4 Parallel SMF
ACC_100G_ACC_25GAUI_C2M_BER_5_	100G ACC (Active Copper Cable) or 25GAUI C2M ACC. Providing a worst BER of $5 * 10^{-5}$
CR4_100G_BASE_25G_BASE_CR_CA_2	100GBASE-CR4 or 25GBASE-CR CA-L
CR_25G_BASE_CA_25G_S_50G_BASE_	25GBASE-CR CA-S
CR_25G_BASE_CA_25G_N_50G_BASE_	25GBASE-CR CA-N
ER4_40G_BASE	40GBASE-ER4
SR_4_X_10G_BASE	4 x 10GBASE-SR
PSM4_PARALLEL_SMF_40G	40G PSM4 Parallel SMF
G959_1_PROFILE_P1I1_2D1	G959.1 profile P1I1-2D1 (10709 MBd, 2km, 1310nm SM)
G959_1_PROFILE_P1S1_2D2	G959.1 profile P1S1-2D2 (10709 MBd, 40km, 1550nm SM)
G959_1_PROFILE_P1L1_2D2	G959.1 profile P1L1-2D2 (10709 MBd, 80km, 1550nm SM)
T_10G_BASE	10GBASE-T with SFI electrical interface
CLR4_100G	100G CLR4
AOC_100G_AOC_25GAUI_C2M_BER_10_	100G AOC or 25GAUI C2M AOC. Providing a worst BER of 10^{-12} or below
ACC_100G_ACC_25GAUI_C2M_BER_10_	100G ACC or 25GAUI C2M ACC. Providing a worst BER of 10^{-12} or below
DWDM2_100GE	100GE-DWDM2 (DWDM transceiver using 2 wavelengths on a 1550nm DWDM grid with a reach up to 80km)

30 sfpStatus (1.4)

30.1 Type Reference

30.1.1 SfpDiagnostics

Real Time Diagnostic for SFP/SFP+/QSFP28/QSFP+, with A2h/A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 9-11) (SFF 8636 Rev2.10 spec Table 6-1)

struct

`temp`	**float** Internally measured module temperature (°C). [96-97], [22-23]
`vcc`	**float** Internally measured supply voltage in transceiver (V). [98-99], [26-27]
`txPwr`	**list** of **float** Measured TX output power (mW). [102-103], [50-57]
`rxPwr`	**list** of **float** Measured RX input power (mW). [104-105], [34-41]

30.1.2 SfpStatus

SFP Status Data Fields, with A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 4-1) QSFP A0h memory space address is specified next to the SFP A0h. (SFF 8636 Rev 2.10 spec Table 6-14)

struct

`identifier`	**sfpConstants.Identifier** Type of transceiver (SFP/SFP+ or unknown). [0],[128]
`transceiver`	**list** of **sfpConstants.Transceiver** Code for electronic or optical compatibility [3-10], [131-138]
`brNominal`	**float** Nominal signalling rate, units of 100 MBd. [12], [140/222]
`vendorName`	**string** SFP vendor name. [20-35], [148-163]
`vendorOUI`	**string** SFP vendor IEEE company ID. [37-39], [165-167]
`vendorPN`	**string** Part number provided by SFP vendor. [40-55], [168-183]
`vendorRev`	**string** Revision level for part number provided by vendor. [56-59], [184-185]
`wavelength`	**string** Laser wavelength. [60-61], [186-187]
`vendorSN`	**string** Serial number provided by SFP vendor. [68-83], [196-211]
`dateCode`	**string** Vendor's manufacturing date code. [84-91], [212-219]
	optional SfpDiagnostics

`diagnostics`	Optional Diagnostics Monitor Data.

31 st2110Types (1.1)

31.1 Type Reference

31.1.1 St2110Standard

The SMPTE ST2110 standards that describe video, audio or ancillary content.

enum

ST2110_20	ST2110 uncompressed video standard
ST2110_22	ST2110 compressed video standard
ST2110_30	ST2110 audio standard
ST2110_31	ST2110 AES3 formatted audio
ST2110_40	ST2110 ancillary standard

32 st2110TypesLabels (1.1)

33 videoUsabilityInfo (1.2)

33.1 Overview

```
# Changelog
## 1.2
- **Added**
  - BT_2100 to `ColorSpace` enum
  - BT_2100 support in `getColourPrimaries()` function
  - BT_2100 support in `getMatrixCoefficients()` function

- **Changed**
  - Corrected typos: BT_2010_SDR to BT_2020_SDR, BT_2011_PQ to BT_2100_PQ, BT_2011_HLG to BT_210
  - Fixed same typos in `getTransferCharacteristics()` function

## 1.1
- **Changed**
  Added BT_601 to `ColorSpace`
  Added BT_601 to `TransferFunc`
```

33.2 Type Reference

33.2.1 ColorSpace

ColorSpace

ColorSpace is a set of colour coordinates for what each of red, green, blue and white "mean"

enum

UNDEFINED	
BT_709	
BT_601	
BT_2020	
BT_2100	

33.2.2 TransferFunc

TransferFunc

Transfer function to convert video signals to light

enum

UNDEFINED	
BT_709	
BT_601	
BT_2020_SDR	
BT_2100_PQ	
BT_2100_HLG	

33.2.3 VuiParameters

VuiParameters

Video Usability Info

struct

`colorSpace`	**ColorSpace** A set of colour coordinates for what each of red, green, blue and white “mean”.
`transfer`	**TransferFunc** Transfer function to convert video signals to light
`matrix`	**ColorSpace** Specifies the matrix coefficients used to derive the luma and chroma from the RGB primaries.